Ayush Maheshwari

Senior Solutions Architect, NVIDIA | Ph.D. [CSE, IIT Bombay] | Foundation Models

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Summary

Ph.D. in ML from CSE, IIT Bombay and Senior Solutions Architect at NVIDIA, working on foundation models, multilingual NLP, Indic LLMs, distributed training, and model evaluation. Previously led Indic LLM development from scratch at Vizzhy, including training on large accelerator cluster.

Experience

NVIDIA **Gurugram, India**
Senior Solutions Architect, NVAITC *Oct 2024 – Present*

- ◇ Working on foundation-model research across multilingual, domain-specific, and scientific AI problems, including data design, model adaptation, evaluation, and deployment on computing platforms.
- ◇ Collaborating with academic researchers, research organizations, and external AI teams on large-scale model development and scientific AI workloads.
- ◇ Reviewing large-scale training and inference setups, and debugging performance issues across data, model, compute, and serving layers.

Vizzhy Inc **Bengaluru, India**
Lead Research Scientist, Language Model Team *Sep 2023 – Sep 2024*

- ◇ Led a 5-member research team that built Indic large language models from scratch, covering data curation, tokenizer training, architecture choices, pretraining, post-training, and evaluation.
- ◇ Trained Indic LLMs on a 1,024 accelerator cluster, handling distributed training, experiment runs, checkpointing, failure recovery, throughput debugging, and model-quality analysis.

Education

Indian Institute of Technology Bombay **Mumbai, India**
Ph.D. in ML *Jan 2019 – Aug 2023*
Defence: July 2024; '**Impactful Research Award 2023**' by IIT Bombay

Thesis: Knowledge Integration in NLP Models using Constraint Ingestion and Generation
Advisor: Prof. Ganesh Ramakrishnan

Tata Institute of Social Sciences **Mumbai, India**
M.Sc. *2013 – 2015*

University College of Engineering **Rajasthan Technical University Kota, India**
B.Tech. (CSE) *2007 – 2011*

Skills

LLM Systems: Foundation Models, Distributed Training, LLM Pretraining, Post-training, Instruction Tuning, Model Adaptation, Inference Optimization and Model Deployment

NLP: Multilingual NLP, Low-Resource Languages, Neural Machine Translation, Semi-Supervised Learning

Programming Languages/Tools: Python, PyTorch, Megatron, Bash, SQL, NumPy, scikit-learn, C, C++, JavaScript

Open-Source contributions

- **SPEAR** : Data-Programming python library to create training data. GitHub
- **UDAAN** : Post-editing workbench to quickly create publishable translations. Website

Additional Experience

Adobe Research — **Bengaluru, India**
Research Intern, Document Cloud Team — *May 2021 – Aug 2021*

NCETIS, IIT Bombay — **Mumbai, India**
Project Engineer — *May 2016 – Dec 2018*

Water & Climate Lab, IIT Gandhinagar — **Gandhinagar, India**
Junior Research Fellow — *May 2015 – Apr 2016*

Tata Consultancy Services — **Mumbai, India**
Systems Engineer — *Oct 2011 – Jun 2013*

Research Interests

- Develop large language models for mid and low-resource languages.
- Develop machine learning based methods that generalize to new domains with limited resources.
- Lexically constrained neural machine translation, data programming, semi-supervised learning

Awards, Services & Talks

- My PhD thesis work was awarded **Impactful Research Award 2023** by IIT Bombay
- Best Demo Paper Award at CODS-COMAD 2023
- Awarded prestigious Ekal Fellowship 2020
- Reviewer: ARR 22–24, EACL, EMNLP, ACL (21–23)
- Fully covered grant for attending EMNLP 2022, Abu Dhabi
- Co-Founder at non-profit Temples of India - A web and app platform listing > 700K temples across Indian subcontinent.
- Key Member of Udaan Project - A project to translate education textbooks in Indian languages.

Computer Skills

Programming Languages: Python, C, C++, Javascript/jQuery, Bash, SQL

Libraries: PyTorch, DeepSpeed, HuggingFace, NumPy, scikit-learn

Talks

- Delivered Masterclass on **Building data-efficient large language models**
  NLTF NASSCOM, 2024
- Invited Tutorial on **Data Efficient Machine Learning for end-to-end Educational Content Creation**
  International Conference on Educational Data Mining, 2023
- Talk on **SPEAR: Data Programming using Semi-Supervision** [Talk]
  DECILE Workshop, 2022

Publications

1. AQBENCH: Benchmarking Neural Surrogates for Air Quality Forecasting

NeurIPS (ED Track), 2026
Siddharthan D, Bedi S, Parmar P, **Maheshwari A**, Kota S, Krishnan N

2. IndicParam: Benchmark to evaluate LLMs on low-resource Indic languages
EMNLP (Main), 2026
Maheshwari, A, Sharma K, Patel V, Maheshwari A

3. LexGen: Domain-aware Multilingual Lexicon Generation
ACL (Main) 2025
Maheshwari, A, Singh A, Karthika NJ, Bhat K, Jyothi P, Ramakrishnan G

4. ARISE: Iterative Rule Induction and Synthetic Data Generation for Text Classification
Findings of NAACL, 2025
Yashwanth M, Singh V, **Maheshwari, A**, Krishna A, Ramakrishnan G

5. DictDis: Dictionary Constrained Disambiguation for Improved NMT
Findings of EMNLP, 2024
Maheshwari, A, Jyothi P, Ramakrishnan G

6. FAIR: Filtering of Automatically Induced Rules
EACL 2024
Bajpai, D.J, **Maheshwari, A**, Hanawal, M. and Ramakrishnan, G

7. Samayik: A Benchmark and Dataset for English-Sanskrit Translation
LREC-COLING 2024
Maheshwari, A, Gupta, A, Krishna, A, Singh, A.K, Ramakrishnan, G, Gourishetty, A.K. and Singla, J

8. Adaptive mixing of auxiliary losses in supervised learning
AAAI, 2023 (Oral Presentation)
Sivasubramanian D, **Maheshwari, A**, Prathosh AP, Shenoy P, Ramakrishnan G

9. UDAAN-Machine Learning based Post-Editing tool for Document Translation
CODS-COMAD Demonstrations, 2023 (Best Demo Paper Award)
Maheshwari, A, R. Ajay, Subramanian V, Ramakrishnan G

10. Eigen: Expert-Informed Joint Learning Aggregation for High-Fidelity Information Extraction from Document Images
Machine Learning for Health (ML4H) at NeurIPS 2023
Singh, A, Venkatapathy, **Maheshwari, A**, Narayan, P, Shetty, D.P. and Ramakrishnan, G

11. Learning to Robustly Aggregate Labeling Functions for Semi-supervised Data Programming
Findings of ACL, 2022
Maheshwari, A, Killamsetty K, Ramakrishnan G, Iyer R, Danilevsky M, Popa L

12. A Benchmark and Dataset for Post-OCR text correction in Sanskrit
Findings of EMNLP, 2022
Maheshwari, A, Singh, N, Krishna, A. and Ramakrishnan G,

13. Data Programming using Semi-Supervision and Subset Selection
Findings of ACL, 2021
Maheshwari, A, Chatterjee O, Killamsetty K, Iyer R, Ramakrishnan G

14. Joint Learning of Hyperbolic Label Embeddings for Hierarchical Multi-label Classification
EACL 2021
Maheshwari A, Chatterjee S, Ramakrishnan, G. Jagaralpudi S.

15. OCR On-the-Go: Robust End-to-end Systems for Reading License Plates & Street Signs
International Conference on Document Analysis and Recognition, 2019
Saluja, R, **Maheshwari, A**, Ramakrishnan, G, Chaudhuri, P. and Carman, M

16. DynGAN: Generative Adversarial Networks for Dynamic Network Embedding
Graph Representation Learning Workshop at NeurIPS 2019
Maheshwari, A, Goyal, A, Hanawal, M.K. and Ramakrishnan, G

17. Semi-supervised data programming in python
EMNLP 2022 - System Demonstrations
Abhishek, G, Ingole, H, Laturia, P, Dorna, V, **Maheshwari, A**, Ramakrishnan, G. and Iyer, R,

18. Rule Augmented Unsupervised Constituency Parsing
Findings of ACL, 2021
Sahay, A, Nasery, A, **Maheshwari, A**, Ramakrishnan, G. and Iyer, R

19. Representation learning on graphs by integrating content and structure information
11th International Conference on Communication Systems & Networks (COMSNETS)
Maheshwari, A, Goyal, A, Kumar, A, Hanawal, M.K. and Ramakrishnan, G

20. Enhancing Low-Resource NMT with a Multilingual Encoder and Knowledge Distillation: A Case Study
LoResMT Workshop at ACL 2024
Roy, A, Ray, P, **Maheshwari, A**, Sarkar, S. and Goyal, P

21. Unsupervised learning of explainable parse trees for improved generalisation
IJCNN 2021
Sahay, A, **Maheshwari, A**, Kumar, R, Ramakrishnan, G, Hanawal, M.K. and Arya, K

22. Tale of tails using rule augmented sequence labeling for event extraction
Star-AI Workshop at AAAI 2019
Maheshwari, A, Patel, H, Rathod, N, Kumar, R, Ramakrishnan, G. and Bhattacharyya, P

23. Entity Resolution and Location Disambiguation in the Ancient Hindu Temples Domain using Web Data
NAACL 2018 - System Demonstrations
Maheshwari, A, Kumar, V, Ramakrishnan, G. and Nath, J.S